

$$(b) x e^{y/x} y' = y e^{y/x} + x$$

← slide #31 (2014/2015)
a)

$$y' = \frac{y e^{y/x} + x}{x e^{y/x}}, \quad x \neq 0 \quad \text{--- (A)}$$

$$f(x, y) = \frac{y e^{y/x} + x}{x e^{y/x}}$$

$$f(\lambda x, \lambda y) = \frac{\lambda y e^{\frac{\lambda y}{\lambda x}} + \lambda x}{\lambda x e^{\lambda y / \lambda x}}$$

$$f(\lambda x, \lambda y) = \frac{y e^{y/x} + x}{x e^{y/x}}, \quad f \text{ é homogênea de grau zero.}$$

Logo, uso a substituição

$$y = z x \Rightarrow z = \frac{y}{x} \Rightarrow z' =$$

$$y' = z' x + z x' = z + x z'$$

Substituo em (A):

$$z + x z' = \frac{z x e^{\frac{z x}{x}} + x}{x e^{\frac{z x}{x}}}, \quad x \neq 0$$

$$z + x z' = \frac{z x e^z + x}{x e^z}$$

$$z + x z' = \frac{z e^z + 1}{e^z}, \quad \text{eg.}$$

$$z + xz' = z + e^{-z}$$

$$xz' = e^{-z}$$

$$z' = \frac{e^{-z}}{x}$$

$$z' = \frac{\frac{1}{x}}{e^z}, \text{ eq. de var. sep.}$$

$$e^z z' - \frac{1}{x} = 0$$

$$e^z \frac{dz}{dx} - \frac{1}{x} = 0$$

$$e^z dz = \frac{1}{x} dx$$

$$\int e^z dz = \int \frac{1}{x} dx$$

$$e^z + C_1 = \ln|x| + C_2, \quad C_1, C_2 \in \mathbb{R}$$

$$e^z = \ln|x| + C_3, \quad C_3 \in \mathbb{R}$$

$$z = \ln(\ln|x| + C), \quad C \in \mathbb{R}_{//}$$

$$(c) (x^3 + y^3) dx - 3y^2 x dy = 0$$

$$(x^3 + y^3) - 3y^2 x \frac{dy}{dx} = 0$$

$$-3y^2 x \frac{dy}{dx} = -(x^3 + y^3)$$

$$\boxed{\frac{dy}{dx} = \frac{x^3 + y^3}{3y^2 x}} \quad \text{--- (A)}$$

$$f(x, y) = \frac{x^3 + y^3}{3y^2 x}$$

$$f(\lambda x, \lambda y) = \frac{(\lambda x)^3 + (\lambda y)^3}{3(\lambda y)^2 (\lambda x)} = \frac{\lambda^3 x^3 + \lambda^3 y^3}{3\lambda^2 y^2 (\lambda x)}$$

$$f(\lambda x, \lambda y) = f(x, y) \Rightarrow f \text{ é homogênea de grau zero}$$

Substituição $y = zx \Rightarrow z = \frac{y}{x}$

$$y' = z'x + zx' \Rightarrow y' = z'x + z$$

Substituo em (A):

$$z + xz' = \frac{x^3 + z^3 x^3}{3z^2 x^2 \cdot x}$$

$$z + xz' = \frac{x^3(1 + z^3)}{x^3(3z^2)}$$

$$z + \lambda z' = \frac{1+z^3}{3z^2}$$

$$\lambda z' = \frac{1+z^3}{3z^2} - z$$

$$\lambda z' = \frac{1+z^3-3z^3}{3z^2}$$

$$\lambda z' = \frac{1-2z^3}{3z^2}$$

$$z' = \frac{\frac{1-2z^3}{3z^2}}{\lambda}$$

$$z' = \frac{\frac{1}{\lambda} \overset{p(\lambda)}{}}{\frac{3z^2}{1-2z^3} \overset{Q(z)}}{}$$

, eg. de var. sep.

$$\frac{3z^2}{1-2z^3} z' - \frac{1}{\lambda} = 0$$

$$\frac{3z^2}{1-2z^3} \frac{dz}{d\lambda} - \frac{1}{\lambda} = 0$$

$$\frac{3z^2}{1-2z^3} dz = \frac{1}{\lambda} d\lambda$$

$$\int \frac{3z^2}{1-2z^3} dz = \int \frac{1}{\lambda} d\lambda = \ln|\lambda| + C_1, C_1 \in \mathbb{R}$$

⌞ (A)

Calculo agora

$$I = \int \frac{3z^2}{1-2z^3} dz$$

$$I = -\frac{1}{2} \int \frac{6z^2}{2z^3-1} dz$$

$$I = -\frac{1}{2} \ln |2z^3-1| + C_2, \quad C_2 \in \mathbb{R}$$

Substitua em (A):

$$-\frac{1}{2} \ln |2z^3-1| + C_2 = \ln |x| + C_1, \quad C_1, C_2 \in \mathbb{R}$$

$$\frac{1}{2} \ln |2z^3-1| + \ln |x| = C, \quad C \in \mathbb{R}$$

$$\ln |2z^3-1| - 2C_2 = -2 \ln |x| - 2C_1$$

$$\ln |2z^3-1| + C_3 = \ln \left(\frac{1}{\sqrt{x}} \right) + C_4, \quad C_3, C_4 \in \mathbb{R}$$

$$e^{\ln |2z^3-1| + C_3} = e^{\ln \left(\frac{1}{\sqrt{x}} \right) + C_4}$$

$$|2z^3-1| e^{C_3} = e^{\frac{1}{\sqrt{x}}} e^{C_4}$$

$$|2z^3-1| = \left(\frac{e^{C_4}}{e^{C_3}} \right)^{\frac{1}{\sqrt{x}}} = \frac{1}{\sqrt{x}}, \quad C_5 \in \mathbb{R}^+$$

$$|2z^3 - 1| = C_5 \frac{1}{\sqrt{x}}, \quad C_5 \in \mathbb{R}^+$$

que é o mesmo que

$$|2z^3 - 1| = \frac{C_6}{\sqrt{x}}, \quad C_6 \in \mathbb{R} \setminus \{0\}$$

$$2 \frac{y^3}{x^3} - 1 = \frac{C_6}{\sqrt{x}}$$

$$\frac{2y^3}{x^3} = \frac{C_6}{\sqrt{x}} + 1$$

$$2y^3 = x^3 \left(\frac{C_6}{\sqrt{x}} + 1 \right)$$

$$y^3 = x^3 \left(\frac{C}{2\sqrt{x}} + \frac{1}{2} \right), \quad C \in \mathbb{R} \setminus \{0\}$$

$$y = x \sqrt{\frac{C}{2\sqrt{x}} + \frac{1}{2}}, \quad C \in \mathbb{R} \setminus \{0\}$$

$$(d) (x+y) dx + (y-x) dy = 0$$

$$x+y + (y-x) \frac{dy}{dx} = 0$$

$$(y-x) \frac{dy}{dx} = -(x+y)$$

$$\boxed{\frac{dy}{dx} = \frac{x+y}{x-y}} \quad \text{--- (A)}$$

$$f(x,y) = \frac{x+y}{x-y}$$

$$f(\lambda x, \lambda y) = \frac{\lambda x + \lambda y}{\lambda x - \lambda y} = f(x,y)$$

f é homogênea de grau zero

Substituição: $y = z x$ em (A)

$$y' = z x' + z' x = z + x z'$$

$$z + x z' = \frac{xz + x}{xz - x}, \quad x \neq 0$$

$$z + x z' = \frac{x(1+z)}{x(1-z)}$$

$$z + x z' = \frac{1+z}{1-z}$$

$$x z' = \frac{1+z}{1-z} - z$$

$$x z' = \frac{1+z}{1-z} - \frac{z(1-z)}{1-z}$$

$$x z' = \frac{1+z-z+z^2}{1-z}$$

$$xz' = \frac{1+z^2}{1-z}$$

$$z' = \frac{\frac{1+z^2}{1-z}}{x}$$

$$z' = \frac{\frac{1}{x}}{\frac{1-z}{1+z^2}}, \text{ eq. var. xp.}$$

$$\frac{1-z}{1+z^2} z' = \frac{1}{x}$$

$$\frac{1-z}{1+z^2} dz = \frac{1}{x} dx$$

$$\int \frac{1-z}{1+z^2} dz = \int \frac{1}{x} dx = \ln|x| + C_1, C_1 \in \mathbb{R}$$

(A)

Calculo agora

$$I = \int \frac{1-z}{1+z^2} dz$$

$$I = \int \frac{1}{1+z^2} dz - \int \frac{z}{1+z^2} dz$$

$$I = \arctg(z) - \frac{1}{2} \ln(1+z^2) + C_2, C_2 \in \mathbb{R}$$

substitue in (A)

$$\operatorname{arctg}(z) - \frac{1}{2} \ln(1+z^2) + C_2 = \ln|z| + C_1, \quad C_1, C_2 \in \mathbb{R}$$

$$\operatorname{arctg}\left(\frac{y}{x}\right) - \frac{1}{2} \ln\left(1+\left(\frac{y}{x}\right)^2\right) - \ln|x| = C, \quad C \in \mathbb{R}$$

$$\operatorname{arctg}\left(\frac{y}{x}\right) - \ln\left(|x| \sqrt{1+\frac{y^2}{x^2}}\right) = C, \quad C \in \mathbb{R}$$

$$\operatorname{arctg}\left(\frac{y}{x}\right) - \ln\left(|x| \sqrt{\frac{x^2+y^2}{x^2}}\right) = C$$

$$\operatorname{arctg}\left(\frac{y}{x}\right) - \ln \sqrt{x^2+y^2} = C, \quad C \in \mathbb{R}$$